

Predicting Football Player Performance Decline After Transfer

DS Bootcamp Capstone – Checkpoint 5

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Why Predict Transfer Decline?

Football clubs invest millions in player transfers.

Many signings underperform after joining new clubs.

Goal: Predict the likelihood of a player's **performance decline** post-transfer.

**Key
Question:**

Given pre-transfer data, can we predict if a player's performance will drop after a move?

Stakeholders: Scouts | Recruiters | Club Managers | Analysts

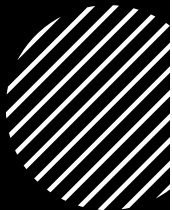
Why It Matters?

- Data-driven recruitment reduces financial risk.
- Objective: identify high-risk transfers before committing resources.
- **Key Metrics:**
 - Accuracy = 0.72
 - ROC-AUC = 0.76
 - F1 (Decline Class) = 0.59
- **Decision Use:** Helps prioritize scouting budgets and optimize player portfolios.





Data Sources & Overview



FBref – Performance stats (minutes, goals, GA90).



Transfermarkt – Transfer fees, market values, league info.



Merged records: $\approx$ 5000 player-seasons.



Key Variables:

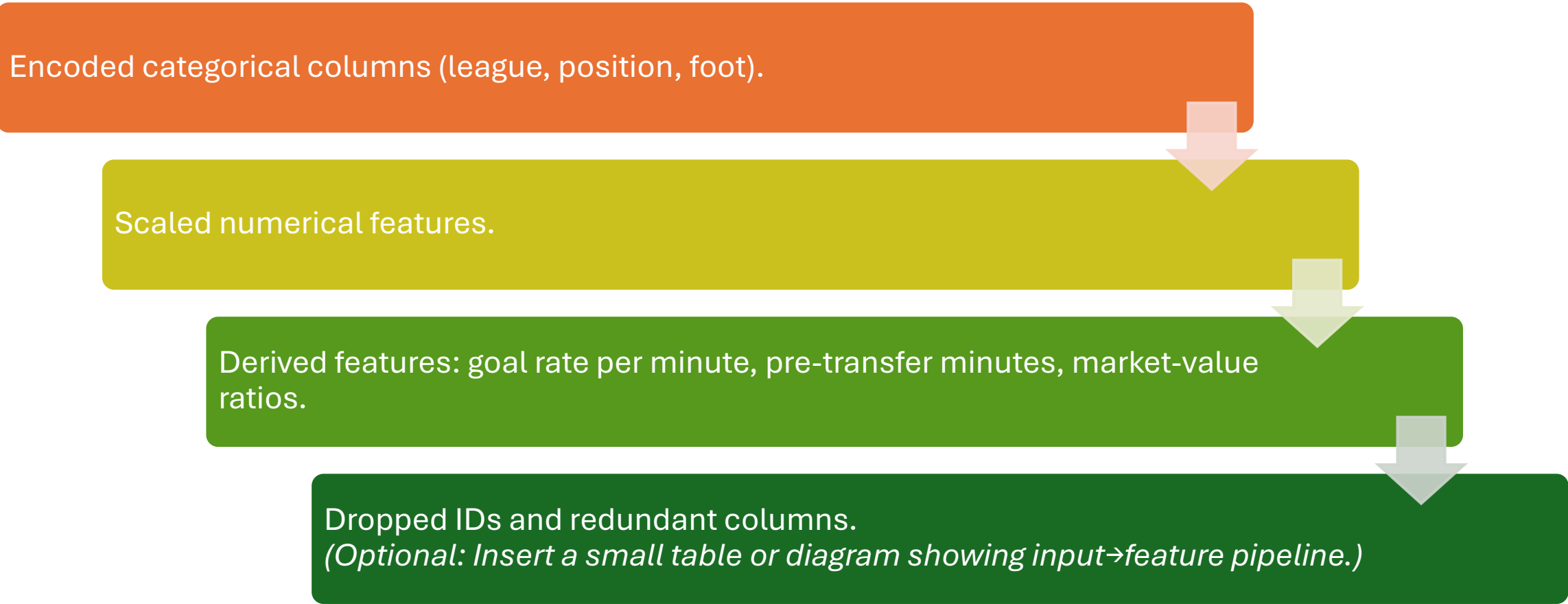
from_league, to_league, position, transfer_fee, market_value_in_eur, height_in_cm, DeclineFlag.



Main Limitation: No injury or morale data.

Turning Raw Data into Signals

Encoded categorical columns (league, position, foot).



```
graph TD; A[Encoded categorical columns (league, position, foot).] --> B[Scaled numerical features.]; B --> C[Derived features: goal rate per minute, pre-transfer minutes, market-value ratios.]; C --> D[Dropped IDs and redundant columns. (Optional: Insert a small table or diagram showing input→feature pipeline.)];
```

Scaled numerical features.

Derived features: goal rate per minute, pre-transfer minutes, market-value ratios.

Dropped IDs and redundant columns.
(Optional: Insert a small table or diagram showing input→feature pipeline.)

From Baselines to Final Model

Model	Notes	ROC-AUC
Logistic Regression	Interpretable baseline	0.70
Decision Tree	Slight overfit	0.72
Random Forest (300 trees)	Best generalization	0.76

Discarded: Gradient Boosting overfit given limited samples.

Results



Final Performance on Test Data



Accuracy = 0.72 ROC-AUC = 0.76 F1 = 0.59
(for Decline class)



Confusion Matrix → show heatmap image
(results/final/confusion_matrix.png)



ROC Curve → show results/final/roc_curve.png



Balanced Precision/Recall between classes.

What the Model Tells Us

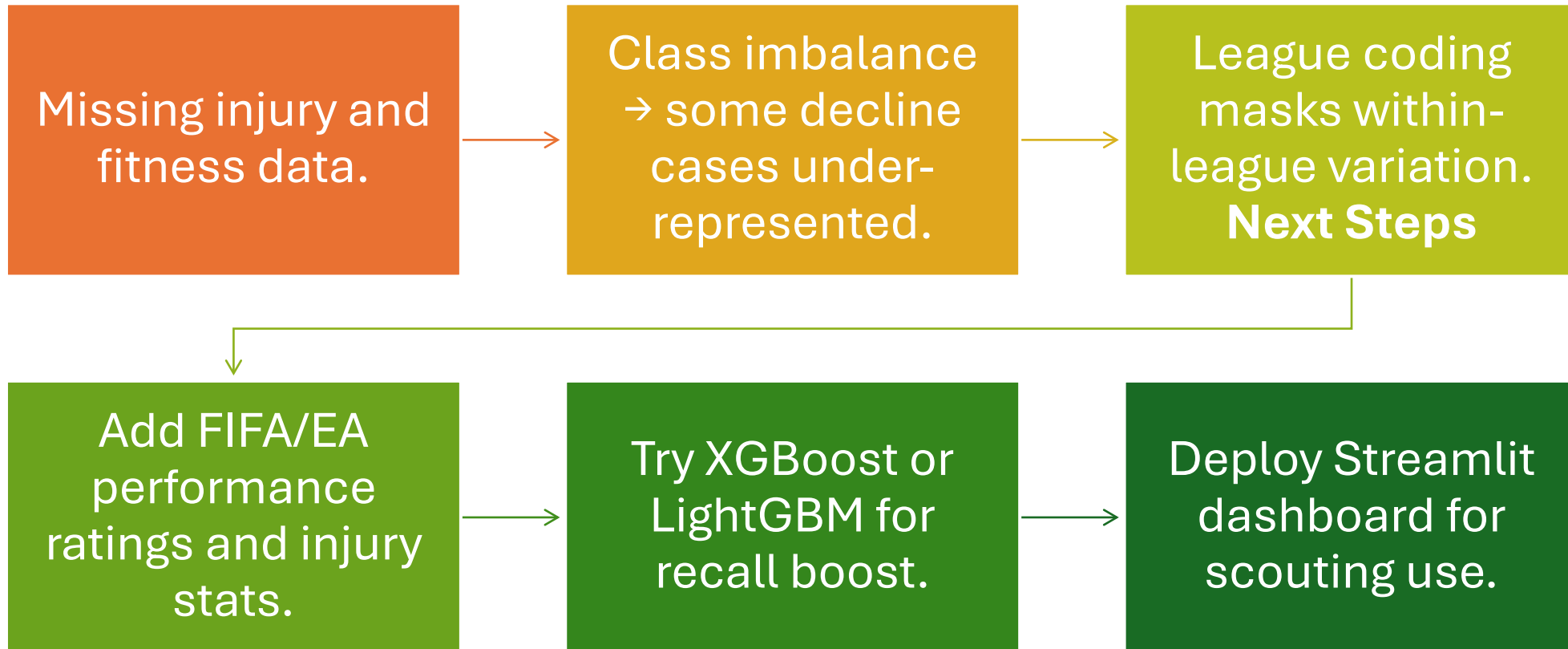
Players moving **to higher-tier leagues** → higher decline risk.

Age, minutes played, and transfer fee are top predictors.

Defenders show more stable performance post-transfer.

The model captures general patterns but not psychological or team fit factors.
(Insert bar chart of top-20 feature importances.)

Limitations & Next Steps



Business Impact & Conclusion

Quantifies risk for
every potential
transfer.

Supports clubs in
contract
negotiations and
budget allocation.

Enables data-
driven scouting to
avoid costly
mistakes.
Final Takeaway:

*A 0.76 AUC model
can meaningfully
reduce transfer
risk through data-
driven analytics.*